



IMAGINE BUILDING DIFFERENTLY

PRODUCT DATA SHEET

52-Wall Assembly B.PUBLIC Prefab

PRODUCT OVERVIEW

Product Description

B.PUBLIC Prefab panelized assemblies exemplify best practices in construction and building science for unparalleled performance, durability, health, and environmental sustainability. The panels are manufactured in controlled environments leveraging both traditional craftsmanship and the latest technology to deliver airtight, vapor-open, and carbon-storing assemblies that achieve Net Zero and Passive House standards. The B.PUBLIC envelope provides industry leading comfort, quiet, and speed to new construction.

Application

The 52-Wall is designed to meet typical type V construction requirements up to 3 stories in the colder IECC climate regions 5, 6, and 7 coast to coast. B.PUBLIC assemblies and component connections minimize **thermally-bridging** allowing for unprecedented design freedom without sacrificing performance. Available in custom widths and heights including custom window and door opening configurations.

Installation

All prefabricated wall assemblies come fully wrapped in rated weather-resistant barriers for durability during delivery and installation. Our unique isolated cavity design coupled with a redundant air-barrier ensures that prefabricated assemblies exceed Passive House air-tightness requirements. Preinstalled hoist pick-points reduce on-site labor and simplify the craning and placing process.

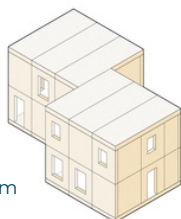
Key Attributes

- Breathable yet water-proof vapor-open assembly
- Class A flame and smoke resistant materials
- Sealed to eliminate energy loss due to wind washing
- Grade 1 highest quality installation rating
- Industry leading acoustic performance
- Minimized thermal bridging per PHI standards
- Biogenic materials / Carbon sequestering / Low embodied energy
- > 70% post consumer recycled content
- Living Building Challenge compliant / Red List free
- Urea-Formaldehyde & foam free / No dangerous off-gassing



TECHNICAL OVERVIEW

Description	Off-site prefabricated light wood frame wall assembly
R Value	14" Cellulose / R-3.7 per inch
Vapor Control	Class II / vapor variable membrane 0.13 to >13.0 perms Rated WRB- 2 months exposure
Cavity Insulation	Dense-pack Cellulose / Borate treated for fire / mold / insect resistance.
Weather Barrier	Fully adhered monolithic non-porous vapor-open weather barrier / rated 6 months exposure.
Thickness	14 1/2" (16" plus 1 1/2" furring for interior service cavity)



B.PUBLIC Prefab PBC
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B.PUBLIC is a woman-owned public benefit corporation. We're a team of architects, designers, tradespeople, and change makers meeting the climate and housing crises head-on. We're reimagining building for the better with technology that achieves extreme energy reduction, deep sustainability, and unmatched livability and comfort.





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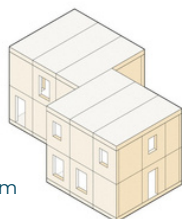
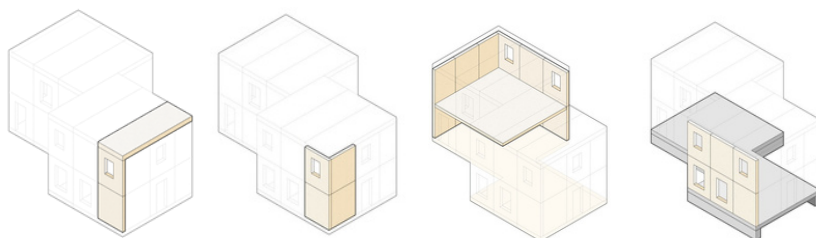
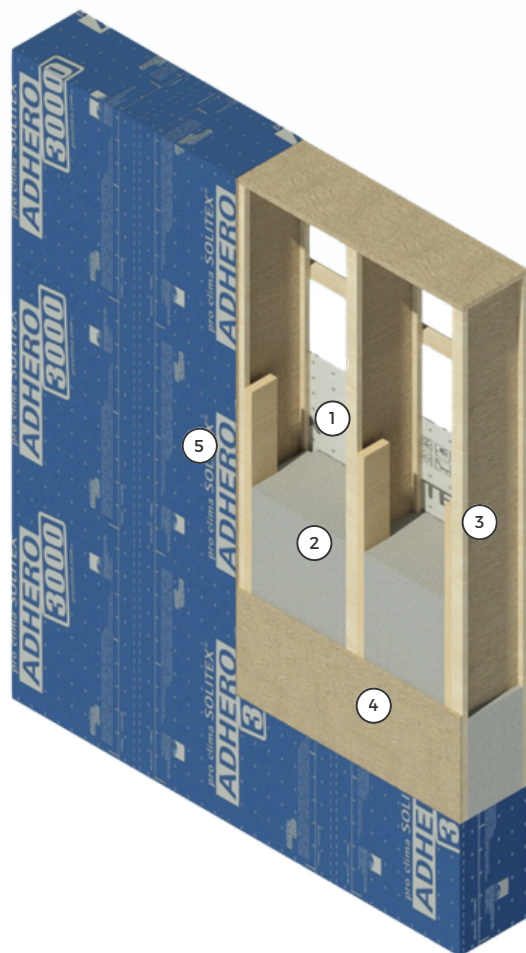
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MATERIAL SPECIFICATIONS

STANDARD ASSEMBLY SPECIFICATIONS*

Description	Off-site prefabricated light wood frame wall assembly
R-Value	14" Cellulose / R-3.7 per inch
Thickness	14 1/2" (16" plus 1 1/2" furring for interior service cavity)
1 Interior Membrane	IBC / IRC Class II vapor retarder Vapor variable membrane 0.13 to >13.0 perms WRB -rated for 2 months exposure Class A Flame spread index 0 Class A Smoke developed index 35 ICC -ES report ESR-4854 Air permeance 0.00004 cfm/ft
2 Cavity Insulation	Dense-pack cellulose >3.5 lb per cubic ft. Borate treated for fire/mold/insect R-3.70 per inch 85% recycled content Formaldehyde free. -35 GWP value kg CO2e Class A fire rated
3 Structure	2x6 SPF studs @ 24" OC with engineered web stud for depth. 1 1/8" Engineered wood plates
4 Sheathing	1/2" plywood sheathing. Glued & nailed
5 Exterior Weather Barrier	Fully adhered monolithic non-porous vapor-permeable weather resistant barrier. Rated 6 months exposure. Self-sealing around nail & screw penetrations. Waterproof to >32ft water column Vapor permanence 11 perms Airtightness 0.00009 cfm/ft2 Class A fire rated FS:10 SDI:15
Dimensional Tolerances	3/16" +/- Width & Height <1/4" out-of-plane <0.3% x high / in-square



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*Subject to change per project engineering and architectural requirements.

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